Product Feature Case Study

Manual Labor Record Management

Reducing errors and support dependency in labor vouchering

Product Context

SmartShop is a shopfloor application used to perform different operations and manage labor vouchering, time tracking, and operational records. Labor data captured in Labor Records module of SmartShop directly impacts payroll, compliance, and production costing.

Existing Workflow

Before this feature, manual labor records (extra hours, downtime, corrections) were not added directly in SmartShop. Instead, the process relied on an external tool

- Labor details were entered manually in SmartSheet.
- SmartSheet allowed free-text entry with no validations
- Records were uploaded and synced into SmartShop
- Support teams manually reviewed and corrected errors before syncing

Resulting Issues

This workflow caused multiple downstream problems:

- Duplicate and overwritten labor records
- Incorrect or missing mandatory fields
- Time overlaps and invalid entries
- High dependency on support teams for corrections
- Low confidence in manually added labor data

Why This Became a Product Problem? – *The issue wasn't just usability, it was data integrity.*

Problem Statement

SmartShop lacked a controlled, validated way for authorized users to add manual labor records. As a result, labor data was entered via external tools without validations, leading to duplicate records, downstream corrections by support teams, and reduced trust in payroll-critical data.

Pain Points:

- No Validation at any point
- Free-text inputs caused inconsistencies
- Duplicate and overlapping records
- Support teams had to manually review and fix entries
- Rework became a routine operational cost
- No Clear Ownership → entered by one → Fixed by another → Approved by a third

How might we?

- Enable authorized users to add manual labor records directly in SmartShop
- Enforce required fields and business rules at the point of entry
- Support both small manual entries and bulk uploads
- Prevent errors without slowing down experienced users
- Eliminate dependency on support teams

Users & Access Control

Primary Users:

- This feature is designed for privileged operational roles, not for all labor users.
- Shopfloor Admins
- Production Supervisors / Managers

Why Access is Restricted?

Manual labor record entry directly affects:

- *Payroll calculations*
- *Compliance and audit trails*
- *Production costing*

Access Control Design:

Manual labor record entry is restricted to authorized admins and supervisors, ensuring accountability while preventing misuse and maintaining payroll-grade data integrity.

Goals & Constraints

Product Feature Goals:

- Reduce errors by adding progressive drop-down inputs
- Validate the input before it gets populated into the tables
- Shift responsibility to authorized users
- Enable bulk import for large datasets
- Prevent duplicates and overlaps
- Use a single screen, consistent review and submission flow

User Experience Goals

- *Fast, repeatable entry for power users*
- *Clear feedback during validation*
- *Ability to review and fix mistakes before submission*
- *No accidental commits*

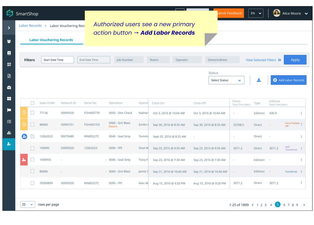
Technical Feasibility:

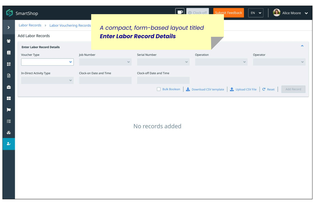
- Manual entry and CSV upload operate on different validation pipelines
- Mixing both in a single session will cause - Sync issues,Partial validation states, Complex rollback logic

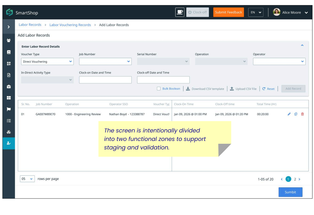
That's why → Design Decision was taken Once one mode is chosen, the other is disabled

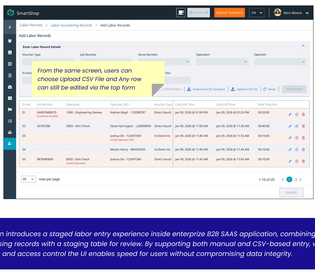
The solution balances speed and control by staging labor records, enforcing validations early, and respecting technical and operational constraints—ensuring accuracy without slowing down experienced users.

Solution Overview









The solution introduces a staged labor entry experience inside enterprize B2B SAAS application, combining a dynamic form for composing records with a staging table for review. By supporting both manual and CSV-based entry, while enforcing validations and access control the UI enables speed for users without compromising data integrity.

Challenges & Key Design Decisions

- Although the requirement sounded straightforward "add labor records manually" this type of workflow had never been implemented within SmartShop before.
  Hence Instead of introducing new components, I reused existing UI elements and re-orchestrated them into a new flow that supports both single and bulk entry without increasing cognitive load.

- A major challenge was making existing components behave in new ways - A form that stages data instead of submitting, A table that acts as a draft workspace,A review step before final commit
  This required careful interaction design to ensure users clearly understood → When data was added,When it was validated,When it was actually submitted

- Defining Validation Boundaries for an MVP → What should be validated instantly? What should be validated only at review? What can realistically be implemented in the first release?
  Hence I added Field-level and dependency validations during entry, Duplicate and consistency checks before review

- Labor vouchering involves many fields, but showing everything upfront would overwhelm users.
  Hence I added Progressive Disclosure of Fields

Outcomes & Impact

- ~60–70% reduction in manual corrections by support teams, as most mandatory field and format errors are now prevented at the point of entry.
- Significant drop in duplicate and overwritten labor records, due to frontend validations and staged review before submission.
- Manual corrections that earlier took hours or days (due to back-and-forth with support) can now be completed within a single session by supervisors.
- Manual labor entry is now handled within SmartShop, eliminating an entire error-prone hand-off step.

Clocking OFF – Thanks!